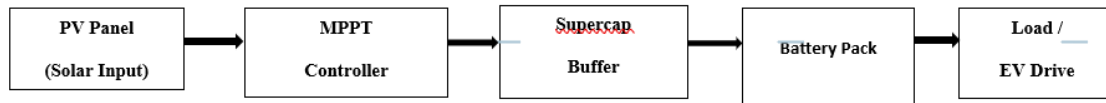


DC-DC Boost Converter Prototype

1. INTRODUCTION

This report includes the work completed during Weeks 3, 4 and 5 of the project. The system requirements were identified and the main components were selected. Design calculations were also carried out to determine the duty cycle, inductor value, capacitor value, and load resistance required for the boost converter.



2. SYSTEM SPECIFICATIONS

Parameter	Value
PV Panel Power	10 W
Open Circuit Voltage (V_{oc})	21.6 V
Maximum Power Voltage (V_{mp})	18 V
Output Voltage (V_{out})	24 V
Switching Frequency	20 kHz
Converter Type	DC-DC Boost Converter

- Solar panel : 10W monocrystalline (305 x 175 mm), $V_{mp} = 18V$, $V_{oc} = 21.6V$
This was selected because its compact size (305 x 175 mm) fits on the small prototype vehicle frame. It is lightweight, affordable, and its V_{mp} of 18V directly provides the required converter input voltage.
- Battery : 24V LiFePO4
LiFePO4 chemistry was selected for its safety, long cycle life (2000+ cycles), and thermal stability. The 24V nominal voltage is greater than the panel V_{mp} of 18V, which confirms that a boost converter is required. At 10W power level, the currents are low and safe for prototype wiring and components.

3. DESIGN CALCULATIONS

Input Voltage (V_{in}) = 18V

Output Voltage (V_{out}) = 24V

Output Power = 100W

Switching Frequency = 20 kHz

4.1 Duty Cycle

For a boost converter:

$$D = 1 - \frac{V_{in}}{V_{out}}$$

$$D = 1 - (18 / 24)$$

$$D = 0.25$$

Required Duty Cycle = 25%

4.2 Input Current

$$I_{in} = P_{out} / V_{in}$$

$$I_{in} = 10 / 18$$

$$I_{in} = 0.56 \text{ A}$$

4.3 Inductor Calculation

Assume ripple current = 30% of input current

$$\Delta I_L = 0.3 \times 0.56$$

$$\Delta I_L = 0.167 \text{ A}$$

Inductor,

$$L = \frac{(V_{in} \times D)}{(\Delta I_L \times f)}$$

$$L = \frac{(18 \times 0.25)}{(0.167 \times 20000)}$$

$$L = 1.347 \text{ mH}$$

Selected value: Inductor = 1.5 mH toroidal inductor

This helps maintain continuous conduction mode (CCM) and reduces current ripple.

4.4 Output Capacitor

Output current:

$$I_{\text{out}} = P_{\text{out}} / V_{\text{out}}$$

$$I_{\text{out}} = 10 / 24$$

$$I_{\text{out}} = 0.417 \text{ A}$$

Allowable ripple voltage:

$$\Delta V_{\text{out}} = 1\% \times 24$$

$$\Delta V_{\text{out}} = 0.24 \text{ V}$$

Capacitor :

$$C = \frac{(I_{\text{out}} \times D)}{(V_{\text{out}} \times f)}$$

$$C = \frac{(0.417 \times 0.25)}{(0.24 \times 20000)}$$

$$C = 21.7 \text{ } \mu\text{F}$$

Selected value:

Capacitor = 47 μF / 50V electrolytic capacitor

A higher capacitor value helps reduce output ripple and improve voltage stability.

Voltage rating of 50V provides adequate safety margin above the 24V output

4.5 Load Resistance

$$R = V_{out}^2 / P_{out}$$

$$R = 24^2 / 10$$

$$R = 57.6 \, \Omega$$

Selected practical value:

Load resistor = 56 Ω or 68 Ω high-power wirewound resistor

4.6 Switching Frequency Selection

Initially, a switching frequency of 50 kHz was considered to reduce passive component size.

However 20 kHz was selected for the prototype because it provides:

- Lower MOSFET switching losses
- Reduced heating
- Easier hardware implementation
- Better stability on breadboard/PCB
- Lower EMI noise
- Easier debugging with oscilloscope.

5. MAIN COMPONENTS SELECTED

Component	Specification
MOSFET	IRFZ44N N-Channel MOSFET
Diode	MBR20100CT Schottky Diode
Inductor	1.35mH Toroidal Inductor
Capacitor	47 μ F / 50V Electrolytic Capacitor
Gate Driver	TLP250 / IR2110
PWM Controller	Arduino / MCU PWM Generation